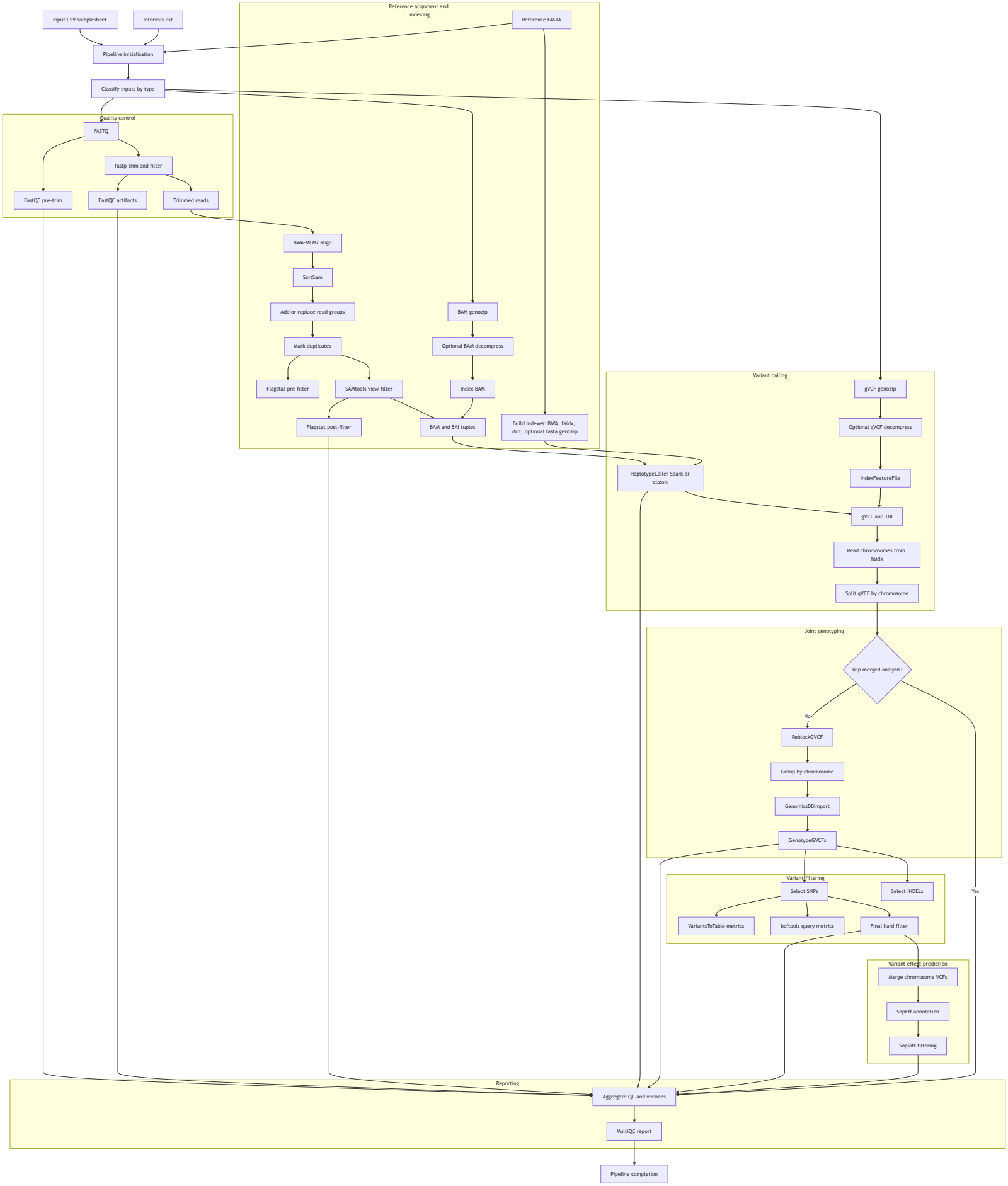
Input CSV samplesheet
Intervals list
Pipeline initialisation
Classify inputs by type
Quality control
FASTQ
fastp trim and filter
FastQC pre-trim
FastQC artifacts
Trimmed reads
Reference alignment and indexing
Reference FASTA
BWA-MEM2 align
SortSam
Add or replace read groups
Mark duplicates
Flagstat pre-filter
SAMtools view filter
Flagstat post-filter
BAM and BAI tuples
BAM genozip
Optional BAM decompress
Index BAM
Build indexes: BWA, faidx, dict, optional fasta genozip
Variant calling
gVCF genozip
Optional gVCF decompress
IndexFeatureFile
gVCF and TBI
Read chromosomes from faidx
Split gVCF by chromosome
HaplotypeCaller Spark or classic
Joint genotyping
skip merged analysis?
No
Yes
ReblockGVCF
Group by chromosome
GenomicsDBImport
GenotypeGVCFs
Variant filtering
Select SNPs
Select INDELs
VariantsToTable metrics
bcftools query metrics
Final hard filter
Variant effect prediction
Merge chromosome VCFs
SnpEff annotation
SnpSift filtering
Reporting
Aggregate QC and versions
MultiQC report
Pipeline completion